

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-CT-7801-07 Sheet 1 of 1
OPL Title	Conductivity / Blowdown Control (AI-7806)	Discipline	Process / Operations	
Equipment	CT-7801 - COOLING TOWER CELL FAN	Area / Unit	7800 - COOLING TOWER SYSTEM	
Related Interlock	SEQ-7801 (SIL 1)	P&ID; Ref	TJC-LLD-PID-7801	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D- S -M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **conductivity / blowdown control (ai-7806)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. High conductivity means the water is over-concentrated; automatic blowdown keeps the cycles of concentration in range. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Read the AI-7806 conductivity value	Permit valid & briefed
2	Verify the automatic blowdown valve XV-7806 operates	Parameter within limit
3	Maintain the target cycles of concentration	Interlock status confirmed
4	Log make-up versus blowdown volumes	Change logged on DCS
5	Coordinate chemical dosing with water treatment	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Routine barscreen cleaning on operator round	Normal debris accumulation	Raked barscreen clear, checked differential level
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised

6. KEY LEARNING POINTS

- High conductivity means the water is over-concentrated; automatic blowdown keeps the cycles of concentration in range.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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